

KCET – 2020

Number of Questions: 60

Total Marks: 120

Chemistry

1. Copper is extracted from copper pyrites by
 - A. Thermal decomposition
 - B. Reduction by coke
 - C. Electrometallurgy
 - D. Auto reduction
2. Function of potassium ethyl xanthate in froth floatation process is to make the ore
 - A. Lighter
 - B. Hydrophobic
 - C. Hydrophilic
 - D. Heavier
3. Sulphide ore on roasting gives a gas X. X reacts with Cl_2 in the presence of activated charcoal to give Y. Y is:
 - A. SO_2Cl_2

- B. S_2Cl_2
C. SCl_6
D. $SOCl_2$
4. Aqueous solution of a salt (A) forms a dense white precipitate with $BaCl_2$ solution. The precipitate dissolves in dilute HCl to produce a gas (B) which decolourises acidified $KMnO_4$ solution A and B respectively are :
- A. $BaSO_3, SO_2$
B. $BaSO_4, H_2S$
C. $BaSO_3, H_2S$
D. $BaSO_4, SO_2$
5. Q5. Bond angle in PH_4^+ is more than that of PH_3 . This is because
- A. Lone pair - bond pair repulsion exists in PH_3
B. PH_4^+ has square planar structure
C. PH_3 has planar trigonal structure
D. Hybridisation of P changes when PH_3 is converted to PH_4^+
6. Incorrectly matched pair is:
- A. XeO_3 pyramidal

- B. XeF_4 tetrahedral
 - C. XeF_6 distorted octahedral
 - D. XeOF_4 square pyramidal
- Q6. Incorrectly matched pair is:

7. Phosphorus pentachloride

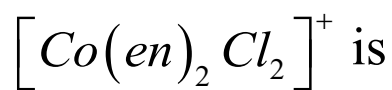
- A. On hydrolysis gives an oxo acid of phosphorus which is tribasic
- B. On hydrolysis gives an oxo acid of phosphorus which is a good reducing agent
- C. Has all the five equivalent bonds
- D. Exists as an ionic solid in which cation has octahedral structure and anion has tetrahedral structure

8. Identify the set of paramagnetic ions among the following:

- A. $\text{V}^{2+}, \text{Co}^{2+}, \text{Ti}^{4+}$
- B. $\text{Ni}^{2+}, \text{Cu}^{2+}, \text{Zn}^{2+}$
- C. $\text{Ti}^{2+}, \text{Cu}^{2+}, \text{Mn}^{3+}$
- D. $\text{Sc}^{3+}, \text{Ti}^{3+}, \text{V}^{3+}$

9. How many moles of acidified $K_2Cr_2O_7$ is required to liberate 6 moles of I_2 from an aqueous solution of I^- ?
- 2
 - 1
 - 0.25
 - 0.5
10. Cu_2Cl_2 and $CuCl_2$ in aqueous medium
- $CuCl_2$ is more stable than Cu_2Cl_2
 - Stability of Cu_2Cl_2 is equal to stability of $CuCl_2$
 - Both are unstable
 - Cu_2Cl_2 is more stable than $CuCl_2$
11. The Co-ordination number of Fe and Co in the complex ions, $[Fe(C_2O_4)_3]^{3-}$ and $[Co(SCN)_4]^{2-}$ are respectively:
- 3 and 4
 - 6 and 8
 - 4 and 6
 - 6 and 4

12. Number of stereoisomers exhibited by



- A. 4
- B. 2
- C. 5
- D. 3

13. Give the IUPAC name of $[Pt(NH_3)_4][PtCl_4]$ is

- A. Tetra ammine platinum (o) tetra chlorido platinum (IV)
- B. Tetra ammine palatinate (II) tetra chlorido platinum (II)
- C. Tetra ammine palatinate (o) tetra chlorido platinum (IV)
- D. Tetra ammine platinum (II) tetra chlorido palatinate (II)

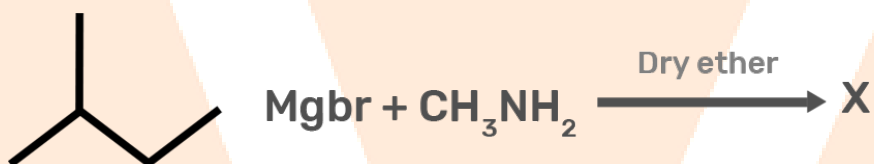
14. Prolonged exposure of chloroform in humans may cause damage to liver. It is due to the formation of the following compound

- A. CCl_4
- B. $COCl_2$
- C. CH_2Cl_2
- D. Cl_2

15. Q15. Which of the following halide shows highest reactivity towards S_N1 reaction?

- A. $C_6H_5CH_2Cl$
- B. CH_3-CH_2Cl
- C. $CH_3-CH_2-CH_2-CH_2I$
- D. C_6H_5Cl

16. In the reaction.



The number of possible isomers for the organic compound X is

- A. 4
- B. 5
- C. 3
- D. 2

17. Which of the following on heating gives an ether as major products?

P: $C_6H_5CH_2Br + CH_3ONa$

Q: $C_6H_5ONa + CH_3Br$

R: $(CH_3)_3C-Cl + CH_3ONa$

S: $C_6H_5CH=CHCl + CH_3ONa$

- A. Both R and S

- B. Both P and R
- C. Both Q and S
- D. Both P and Q

18. The steps involved in the conversion of propan -2-ol to propan -1-ol are in the order

- A. Dehydration, addition of HBr, heating with aq. KOH
- B. Heating with PCl_5 , heating with alc. KOH, acid catalysed addition of water
- C. Heating with PCl_5 , heating with alc. KOH, hydroboration oxidation
- D. Dehydration, addition of HBr in presence of peroxide, heating with alc. KOH

19. Which of the following is the strongest base?

- A. CH_3COO^-
- B. Cl^-
- C. OH^-
- D. CH_3O^-

20. The product 'P' is



- CHO
- A. CH_2CHO
 $\text{CH} = \text{CH} - \text{CH}_3$
- B. $\text{CH} = \text{CH} - \text{CH}_3$
- C. CHO
- D. CH_2CHO

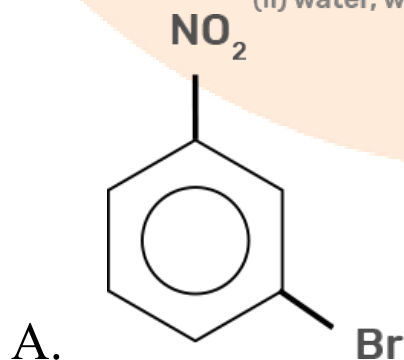
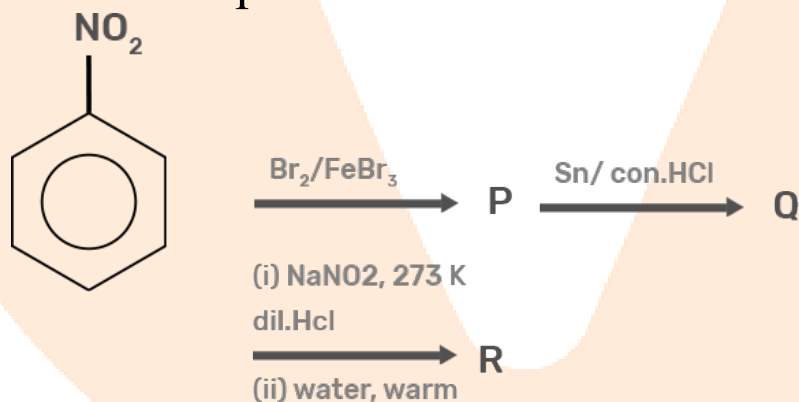
21. Which of the following has the lowest boiling point?

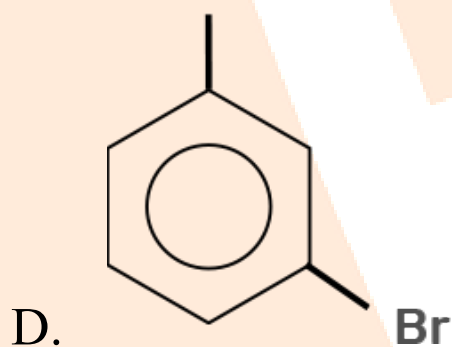
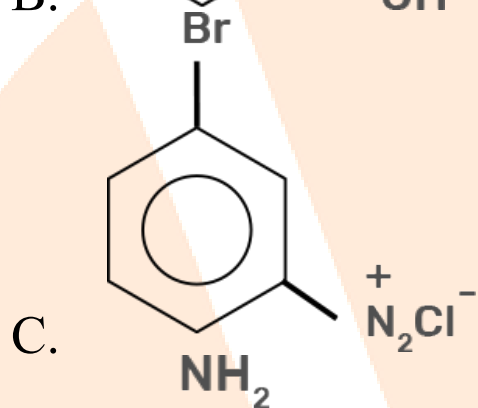
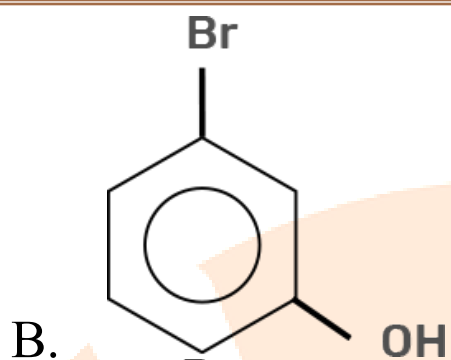
- A. $\text{CH}_3\text{CH}_2\text{OH}$
- B. $\text{CH}_3\text{--CH}_2\text{--NH}_2$
- C. $\text{CH}_3\text{--O--CH}_3$
- D. HCOOH

22. The carbonyl compound that does not undergo aldol condensation is

- A. Acetone
- B. Di chloro acetaldehyde
- C. Tri chloro acetaldehyde
- D. Acetaldehyde

23. The final product is





24. Hinsberg's reagent is

- A. $(\text{CH}_3\text{CO})_2\text{O}$ /pyridine
- B. $\text{C}_6\text{H}_5\text{SO}_2\text{Cl}$
- C. $\text{C}_6\text{H}_5\text{SO}_2\text{NH}_2$
- D. CH_3COCl /pyridine

25. Which one of the following vitamins is not stored in adipose tissue?

- A. A
- B. B₆
- C. D
- D. E

26. Hypothyroidism is caused by the deficiency of

- A. Vitamin B-12
- B. Adrenalin
- C. Thyroxine
- D. Glucocorticoid

27. C₁–C₄ glycosidic bond is NOT found in

- A. Maltose
- B. Sucrose
- C. Lactose
- D. Starch

28. Which of the following polymer has strongest intermolecular forces of attraction?

- A. Neoprene
- B. Terylene
- C. Polythene
- D. Polystyrene

29. Which of the following monomers can undergo condensation polymerization?
- A. Styrene
 - B. Glycine
 - C. isoprene
 - D. Propene
30. A food additive that acts as an antioxidant is
- A. BHA
 - B. Saccharin
 - C. Sugar syrup
 - D. Salt
31. Which of the following is not related to drug-enzyme interaction?
- A. Allosteric site
 - B. Antagonist
 - C. Co-enzymes
 - D. Enzyme inhibitor
32. 0.4 g of dihydrogen is made to react with 7.4 g of dichlorine to form hydrogen chloride. The volume of hydrogen formed at 273K and 1 bar pressure is
- A. 9.08L

- B. 4.54L
- C. 90.8L
- D. 45.4L

33. With regard to photoelectric effect, identify the correct statement among the following

- A. Energy of e^- ejected increases with the increase in the intensity of incident light
- B. Number of e^- ejected increases with the increase in the frequency of incident light
- C. Number of e^- ejected increases with the increase in work function
- D. Number of e^- ejected increases with the increase in the intensity of incident light

34. The last element of the p-block in 6th period is represented by the outer most electronic configuration

- A. $7s^2 7p^6$
- B. $5f^{14} 6d^{10} 7s^2 7p^5$
- C. $4f^{14} 5d^{10} 6s^2 6p^4$
- D. $4f^{14} 5d^{10} 6s^2 6p^6$

35. The conjugate base of NH_3 is

- A. NH_4^+
- B. NH_4OH
- C. NH_2OH
- D. NH_2^-

36. A gas mixture contains 25% He and 75% CH_4 by volume at a given temperature and pressure. The percentage by mass of methane in the mixture is approximately _____

- A. 75%
- B. 25%
- C. 92%
- D. 8%

37. The percentage of s-character in the hybrid orbitals of nitrogen in NO_2^+ , NO_3^- respectively are

- A. 33.3%, 50%, 25%
- B. 33.3%, 25%, 50%
- C. 50%, 33.3%, 25%
- D. 25%, 50%, 33.3%

38. The formal charge on central oxygen atom in ozone is

- A. -1
- B. 0
- C. +2
- D. +1

39. When the same quantity of heat is absorbed by a system at two different temperatures T_1 and T_2 , such that $T_1 > T_2$, change in entropies are ΔS_1 and ΔS_2 respectively. Then

- A. $\Delta S_1 < \Delta S_2$
- B. $\Delta S_1 = \Delta S_2$
- C. $S_1 > S_2$
- D. $\Delta S_2 < \Delta S_1$

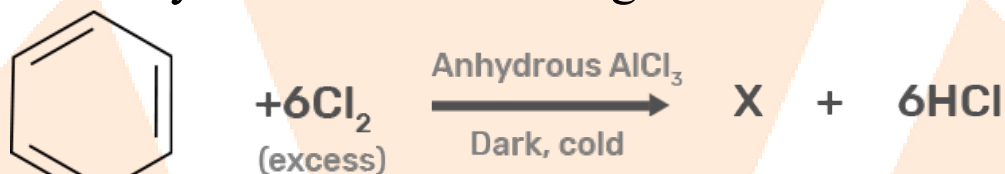
40. The oxidation number of nitrogen atoms in NH_4NO_3 are

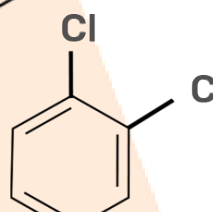
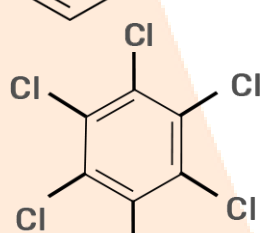
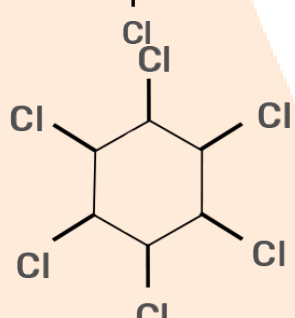
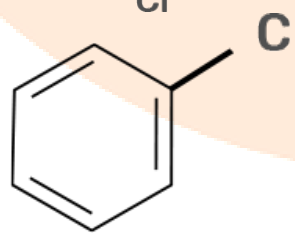
- A. +5, +5
- B. -3, +5
- C. +3, -5
- D. -3, -3

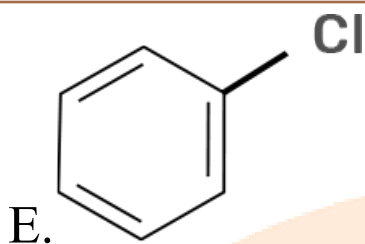
41. A Lewis acid 'X' reacts with LiAlH_4 in ether medium to give a highly toxic gas. This gas when heated with NH_3 gives a compound commonly known as inorganic benzene. The gas is
- B_2O_3
 - B_2H_6
 - $\text{B}_3\text{N}_3\text{H}_6$
 - BF_3
42. The oxide of potassium that does not exist is
- K_2O
 - KO_2
 - K_2O_2
 - K_2O_3
43. The metal that produces H_2 with both dil HCl and NaOH(aq) is
- Zn
 - Mg
 - Ca
 - Fe
44. Which of the following is NOT a pair of functional isomers?
- $\text{C}_2\text{H}_5\text{OC}_2\text{H}_5$ and $\text{C}_3\text{H}_7\text{OCH}_3$

- B. $\text{CH}_3\text{CH}_2\text{OH}$ and CH_3OCH_3
 C. $\text{CH}_3\text{CH}_2\text{NO}_2$ and $\text{H}_2\text{NCH}_2\text{COOH}$
 D. CH_3COOH and HCOOCH_3

45. Identify 'X' in the following reaction



- A. 
- B. 
- C. 
- D. 



46. Which of the following is NOT a green house gas?

- A. CFC
- B. CO_2
- C. O_2
- D. NO_2

47. A metal exists as an oxide with formula $\text{M}_{0.96}\text{O}$.

Metal M can exist as M^{+2} and M^{+3} in its oxide

$\text{M}_{0.96}\text{O}$. The percentage of M^{+3} in the oxide is nearly

- A. 8.3%
- B. 4.6%
- C. 5%
- D. 9.6%

48. A metal crystallises in face centred cubic structure with metallic radius $\sqrt{2}A^\circ$. The volume of the unit cell (in m^3) is

- A. 4×10^{-10}
- B. 6.4×10^{-29}
- C. 4×10^{-9}

D. 6.4×10^{-30}

49. Silicon doped with gallium forms

- A. n-type semiconductor
- B. both n and p type semiconductor
- C. an intrinsic semiconductor
- D. p-type semiconductor

50. The pair of electrolytes that possess same value for the constant (A) in the Debye - Huckel - Onsager equation, $\lambda_m = \lambda_m^e - A\sqrt{C}$ is

- A. $\text{MgSO}_4, \text{NaSO}_4$
- B. $\text{NH}_4\text{Cl}, \text{NaBr}$
- C. $\text{NaBr}, \text{MgSO}_4$
- D. $\text{NaCl}, \text{CaCl}_2$

51. Which of the following pair of solutions is isotonic ?

- A. 0.01M BaCl_2 and 0.015M NaCl
- B. $0.001\text{M Al}_2(\text{SO}_4)_3$ and 0.01M BaCl_2
- C. 0.001M CaCl_2 and $0.001\text{M Al}_2(\text{SO}_4)_3$
- D. 0.01M BaCl_2 and 0.001M CaCl_2

52. Solute 'X' dimerises in water to the extent of 80%.

2.5g of 'X' in 100g of water increases the boiling point by 0.3°C . The molar mass of 'X' is

$$\left[K_b = 0.52 \text{ K kg mol}^{-1} \right]$$

A. 13

B. 52

C. 65

D. 26

53. Given $E_{\text{Fe}^{+3}/\text{Fe}^{+2}}^0 = +0.76\text{V}$ and $E_{\text{I}_2/\text{I}^-}^0 = +0.55\text{V}$. The equilibrium constant for the reaction taking place in galvanic cell consisting of above two electrodes is

$$\left[\frac{2.303RT}{F} = 0.06 \right]$$

A. 1×10^7

B. 1×10^9

C. 3×10^8

D. 5×10^{12}

54. If an aqueous solution of NaF is electrolyzed between inert electrodes, the product obtained at anode is

A. F_2

B. H_2

C. Na

D. O₂

55. In which of the following cases a chemical reaction is possible ?

A. ZnSO_{4(aq)} is placed in a copper vessel

B. AgNO₃ solution is stirred with a copper spoon

C. Conc. HNO₃ is stored in a platinum vessel

D. gold ornaments are washed with dil HCl

56. The time required for 60% completion of a first order reaction is 50 min. The time required for 93.6% completion of the same reaction will be

A. 100 min

B. 83.8 min

C. 50 min

D. 150 min

57. For an elementary reaction $2A + 3B \rightarrow 4C + D$ the rate of appearance of C at time 't' is $2.8 \times 10^{-3} \text{ mol L}^{-1} \text{ S}^{-1}$. Rate of disappearance of B at 't' t will be

A. $\frac{4}{3} (2.8 \times 10^{-3}) \text{ mol L}^{-1} \text{ S}^{-1}$

B. $\frac{3}{4} (2.8 \times 10^{-3}) \text{ mol L}^{-1} \text{ S}^{-1}$

- C. $\frac{3}{4}(2.8 \times 10^{-3}) \text{ mol L}^{-1} \text{ s}^{-1}$
- D. $\frac{1}{4}(2.8 \times 10^{-3}) \text{ mol L}^{-1} \text{ s}^{-1}$

58. The rate constant of a reaction is given by $k = PZ e^{-E_a/RT}$ under standard notation. In order to speed up the reaction, which of the following factors has to be decreased ?

- A. Z
- B. Both Z and T
- C. E_a
- D. T

59. A sol of AgI is prepared by mixing equal volumes of 0.1M AgNO_3 and 0.2M KI, which of the following statement is correct ?

- A. Sol obtained is a negative sol with NO_3^- adsorbed on AgI
- B. Sol obtained is a positive sol with Ag^+ adsorbed on AgI
- C. Sol obtained is a positive sol with K^+ adsorbed on AgI
- D. Sol obtained is a negative sol with I^- adsorbed on AgI

60. During Adsorption of a gas on a solid

- A. $\Delta G < 0, \Delta H < 0, \Delta S < 0$
- B. $\Delta G > 0, \Delta H > 0, \Delta S > 0$
- C. $\Delta G < 0, \Delta H < 0, \Delta S > 0$
- D. $\Delta G < 0, \Delta H > 0, \Delta S < 0$